

PATH 548L Quiz 4 2025W2 June 18, 2026

Quiz rules: There are 20 marks possible.

You can work in teams of 2-3 if you wish and submit a single set of answers with all team member names. Answers to quiz questions must be in narrative, tabular or graphical format as appropriate. Include appropriate captions, legends as appropriate and a short narrative explaining the results.

We want to know the rationale of why you have chosen to address the problem in the way that you have. Although JASP and Python are used in the course, you are free to use any software that you like to answer the questions.

The quiz is **take-home** and consequently open-book with internet access.

Students are permitted to use artificial intelligence tools, including generative AI, to gather information, review concepts or to help produce assignments. However, students are ultimately accountable for the work they submit, and any content generated or supported by an artificial intelligence tool must be cited appropriately. Please be aware of what constitutes academic misconduct at UBC. You can find information here <https://academicintegrity.ubc.ca/regulation-process/academic-misconduct/>.

The JASP info buttons could be an extremely valuable resource for you. If you have difficulty in JASP or python, you can communicate with Andy or Helene on piazza or by email (aroeth@bccrc.ca or Helene.cote@ubc.ca).

Please turn the quiz in as a PDF document in the form **lastname_quiz4.pdf** i.e. **roth_quiz4.pdf**. If submitting as a team, include all last names in alphabetical order separated by an underscore i.e. **cote_roth_quiz4.pdf**. The quiz is due June 25 @1pm by email to both of us (aroeth@bccrc.ca; Helene.cote@ubc.ca)

Please advise Andy and Helene if you experience circumstances that will prevent you from making this deadline.

QUIZ START

Question 1 (6 marks): You have set up a study with 2 exposure arms (treatment A & B) with a binary outcome (develop disease or not). You have 24 mice that you have equally allocated across the two arms. Seventy-five percent of the mice have developed disease in treatment A while 50% did in treatment B. Perform an analysis to determine if the observed differences are statistically significant and report the observed differences in terms of a ratio (with confidence intervals) and interpret that measure. Why did you choose that measure?

Perform a retrospective (post-hoc) power analysis to determine the minimum number of mice you would need to detect significant differences in this study utilizing conventional values used in sample size calculations.

Question 2 (4 marks): Revisiting the dataset for question 3 from quiz 1, show us the correlation between CD19 and CD4 in a figure with appropriate figure legend. The unit of the cell count (is cell per microliter).

Question 3 (10 marks): Revisiting the dataset from question 2 of quiz 2, present in table format the summary statistics for each sex at each time point, as well as the % difference.

Then represent the same in graphical format. Show a graph with lines linking the baseline and follow-up weight for each animal with either colour to differentiate sex or two panels, one for each sex. Next show another graph with the relative weight difference for each sex with appropriate statistics.

- ***Be sure to clearly state the statistical methodology and any assumption checks used.***
- ***Ensure all figures and tables are of publication quality***
- ***Ensure your report is logically structured and easy to follow***